

HDOC Day-15

Question: Menu-Driven Program to Check Even and Prime Digits

1. Problem Statement & Identification

- **Problem:** Create a menu-driven program using switch-case to input a positive integer and perform one of the following operations:
 1. Check whether all digits of the integer are even.
 2. Check whether all digits of the integer are prime numbers (2, 3, 5, 7).
- **Input:** A positive integer (num) and menu choice (choice).
- **Output:** Verification output indicating whether all digits are even or prime.

2. Algorithm

1. **Start**
2. Input a positive integer num.
3. Display menu options (1 and 2).
4. Read user choice.
5. Execute operations based on choice using switch-case:

- **Case 1:**

1. Initialize flag `all_even = 1` and set `temp = num`.

2. While `temp > 0`:

- Extract digit `digit = temp % 10`.
- If `digit % 2 != 0`, set `all_even = 0` and break out of loop.
- Update `temp = temp / 10`.

3. If `all_even == 1`, print "All digits are even", else print "All digits not even".

- **Case 2:**

1. Initialize flag `all_prime = 1` and set `temp = num`.

2. While `temp > 0`:

- Extract digit `digit = temp % 10`.
- If digit is NOT equal to 2, 3, 5, or 7, set `all_prime = 0` and break out of loop.
- Update `temp = temp / 10`.

3. If `all_prime == 1`, print "All digits are prime", else print "All digits are not prime".

- **Default:** Print "Invalid Choice!".

6. End

3. Test Case Table

Test Case	Input Number	Choice	Operation	Expected Output	Actual Output	Status
1	12345	1	Check All Even	All digits not even	All digits not even	Pass
2	24680	1	Check All Even	All digits are even	All digits are even	Pass
3	2357	2	Check All Prime	All digits are prime	All digits are prime	Pass
4	2379	2	Check All Prime	All digits are	All digits are not	Pass

Test Case	Input Number	Choice	Operation	Expected Output	Actual Output	Status
				not prime	prime	

4. C Source Code

```
#include <stdio.h>

int main(void) {
    int choice, num, temp, digit;
    int all_even = 1;
    int all_prime = 1;

    printf("Enter a positive integer: ");
    if (scanf("%d", &num) != 1 || num <= 0) {
        printf("Invalid input! Please enter a positive integer.\n");
        return 1;
    }

    printf("\n--- MENU ---\n");
    printf("1. Check whether all digits are even\n");
    printf("2. Check whether all digits are prime\n");
    printf("Enter your choice (1 or 2): ");
    scanf("%d", &choice);

    switch (choice) {
        case 1:
            temp = num;
            while (temp > 0) {
                digit = temp % 10;
                if (digit % 2 != 0) {
                    all_even = 0;
                    break;
                }
            }

```

```

    }
    temp /= 10;
}

if (all_even) {
    printf("\nNumber: %d, Output: All digits are even\n", num);
} else {
    printf("\nNumber: %d, Output: All digits not even\n", num);
}
break;

case 2:
    temp = num;
    while (temp > 0) {
        digit = temp % 10;
        if (digit != 2 && digit != 3 && digit != 5 && digit != 7) {
            all_prime = 0;
            break;
        }
        temp /= 10;
    }

    if (all_prime) {
        printf("\nNumber: %d, Output: All digits are prime\n", num);
    } else {
        printf("\nNumber: %d, Output: All digits are not prime\n", num);
    }
    break;

default:
    printf("\nInvalid Choice!\n");
    return 1;
}

return 0;
}

```

5. Compilation, Execution & Screenshots

```
kali@kali: ~/HDOOC_Day_15
Session Actions Edit View Help
kali@kali:~$ cd HDOOC_Day_15
kali@kali:~/HDOOC_Day_15$ nano q1.c
kali@kali:~/HDOOC_Day_15$ gcc q1.c -o q1
kali@kali:~/HDOOC_Day_15$ ./q1
Enter a positive integer: 24680

--- MENU ---
1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice (1 or 2): 1

Number: 24680, Output: All digits are even

kali@kali:~/HDOOC_Day_15$
```